

Piotr Krasnowski

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R&D Experience

7/2021 - present **Senior Research Engineer:** *Huawei Paris Research Center, France*

- AI-Native 6G Systems: trustworthy multi-agent systems for 6G wireless networks, Verified AI techniques for agentic reasoning, multi-modal data fusion using distributed Transformers, semantic communications
- Supervising a team and leading R&D projects on Verified AI, multi-agent systems, and distributed ML
- Contributing to 30+ technical inventions and high-impact publications (NeurIPS 2023, ISIT 2024, ICLR 2025)
- Collaborating towards defining 6G standards at international organizations (3GPP, ETSI, IEEE)
- Awards: Best Consultant of Huawei Paris Research Center, Best Patent Contribution, Huawei Future Star

1/2021 - 6/2021 **Research Engineer:** *I3S-CNRS, Sophia Antipolis, France*

- Automatic detection, classification, and analysis of underwater biotic and anthropogenic sounds using Machine Learning, GPU computing, and advanced audio signal processing

12/2017 - 1/2020 **Research Engineer:** *BlackBoxSécu, Sophia Antipolis, France*

- Focus on speech codecs, real-time cryptographic algorithms, error correction coding, system integration

10/2014 - 6/2017 **Research Student:** *R&D Group of Electronics for Spacecrafts, WRUST, Poland*

- ESA's satellite testing, EM compatibility measurements, coding, airborne radar remote sensing

Education

10/2024 - 5/2025 **Management of Technological Innovation**

L'École Polytechnique Executive Education, France

- Certification program on managing innovation projects under time and resource constraints

12/2017 - 12/2020 **Doctor of Philosophy in Computer Science**

I3S-CNRS, The University of Côte d'Azur, France

- Research topics: real-time speech/audio signal processing, digital telecommunications, cryptographic algorithms and protocols, GPU-accelerated Machine Learning on large speech corpora
- Best PhD award of 2021 in Computer Science from l'Ecole Doctorale STIC
- Cooperation with the cybersecurity start-up BlackBoxSécu (Sophia Antipolis, FR)
- Supported and co-funded by the Agence de l'Innovation de Défense and the Ministère des Armées

10/2012 - 9/2017 **Master of Engineering in Telecommunications**

The University of Nottingham, UK • The Wroclaw University of Technology, Poland

10/2012 - 6/2016 **Bachelor of Mathematics**

The Wroclaw University of Science and Technology, Poland

Skills

- Signal processing and telecommunications (wireless/speech, 5G/6G networks, sensing)
- Machine Learning, GPU computing, Tensorflow, PyTorch, Python, MATLAB
- Agentic AI: smolagents/Langchain, architecture design, datasets creation, RL fine-tuning
- Leadership, mentoring, collaboration, and organizational skills
- Patenting (writing Invention Disclosure Forms, collaboration with patent attorneys, etc.)
- Fluent English, advanced French, native Polish

Selected Publications

- Sefidgaran, Milad, Abdellatif Zaidi, and Piotr Krasnowski. "Generalization guarantees for representation learning via data-dependent gaussian mixture priors." *2025 International Conference on Learning Representations (ICLR)* (2025).
- Sefidgaran, Milad, Abdellatif Zaidi, and Piotr Krasnowski. "Minimal communication-cost statistical learning." *2024 IEEE International Symposium on Information Theory* (2024).
- Sefidgaran, Milad, Abdellatif Zaidi, and Piotr Krasnowski. "Minimum Description Length and Generalization Guarantees for Representation Learning." *Advances in Neural Information Processing Systems* (NeurIPS) 36 (2024).
- Krasnowski, Piotr, Jerome Lebrun, and Bruno Martin. "A novel distortion-tolerant speech encryption scheme for secure voice communication." *Speech Communication* 143 (2022)
- Krasnowski, Piotr, Jerome Lebrun, and Bruno Martin. "Introducing a novel data over voice technique for secure voice communication." *Wireless Personal Communications* (2022)